

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built data-backed automation flows that combined public-site API lookups, generated AI-assisted drafts, validation, retries, and operator review before execution.
- Reduced manual filing time by **85%** through backend state checks, failure recovery, and repeatable browser automation.
- Deployed Dockerized services through Google Cloud Run and GitHub Actions, improving iteration speed by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python ML tooling for MRI segmentation across **95 anatomical regions**, covering preprocessing, inference integration, generated masks, and large-array data paths.
- Reduced runtime by **87%** using asynchronous and parallel execution over expensive NumPy/PyTorch-style workflows.
- Validated model outputs through inspectable masks and 3D surfaces while debugging label mapping and data-orientation issues.

Open Source

Contributions

- Contributed maintainer-reviewed patches across **Kuadrant MCP Gateway**, Kyverno, IOOS, Tiled, CRIU, and VideoLAN, with emphasis on reliability, integration behavior, parser robustness, and security-validation paths.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, OpenTelemetry, SQLAlchemy, Typer, Docker*

- Built AI-agent evaluation infrastructure with trace ingestion, evaluator registries, replay modes, failure mining, validation runs, gate decisions, promotions, audit logs, and signed manifests.
- Designed reproducible runs around immutable inputs, content-addressed artifacts, project-scoped API keys, worker leases, JSON schema checks, and latency/cost constraints.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, yfinance, JSONL*

GitHub

- Built a Python multi-agent research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed state, checkpoints, and replayable JSONL traces.
- Evaluated decisions with historical-market backtests, reward-loop scorecards, baseline comparisons, semantic lesson retrieval, and return/drawdown/win-rate signals.

Swish Support System | *JavaScript, React, Python, PostgreSQL/pgvector, Redis, Langfuse*

GitHub

- Built a hybrid AI + rules support system where LLM assessments classify messy complaint semantics while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with PostgreSQL/pgvector, Redis state, Langfuse traces, confidence thresholds, and regression checks.

Technical Skills

Languages: Python, TypeScript, SQL, Go

Frameworks: FastAPI, PyTorch, scikit-learn, React, Next.js, Agno

Tools: NumPy, pandas, PostgreSQL/pgvector, SQLite, Redis, Docker, OpenTelemetry, Langfuse, JSONL